

Seonghwan Park

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EDUCATION

- Ph.D. in Robotics and Mechatronics Engineering**, DGIST, Daegu, South Korea Mar. 2019 – Present (Feb. 2027, expected)
- Integrated M.S./Ph.D. program. Advisor: Prof. Inkyu Moon
 - Dissertation: AI-Driven Intelligent Digital Holographic Microscopy for Automated Phase Reconstruction and Label-Free Quantitative Cellular Analysis
- B.S., School of Undergraduate Studies (Engineering Track)**, DGIST, Daegu, South Korea Mar. 2014 – Feb. 2019

RESEARCH EXPERIENCE

- Researcher**, Research Center of Robotics and Mechatronics Engineering, DGIST Jan. 2021 – Dec. 2024
- Developed deep-learning pipelines for holographic and fluorescence cell imaging, and AI-based cryptanalysis, under national research programs.

PUBLICATIONS

20 journal articles (12 as first author), 8 international conference papers, 1 domestic conference paper. Journal Impact Factors and JCR category ranks are those of the JCR edition current at the time of submission.

Journal Articles — Under Review / In Press

- S. Park**, S. Moon, J. Park, B. Javidi, and I. Moon, "Single-shot physics-informed multimodal reconstruction of quantitative phase and virtual 3D confocal fluorescence from a Gabor hologram", *Laser & Photonics Reviews*, under review (2026) [IF: 9.7] (Q1, top 8.1% in Optics)
- S. Park**, J. Lee, J. Park, and I. Moon, "BiMODiffusion: Physics-aware unsupervised cyclic diffusion for bidirectional virtual fluorescence and phase reconstruction from single-modality imaging", *IEEE Transactions on Multimedia*, under review (2026) [IF: 9.9] (Q1, top 2% in Computer Science, Software Engineering)
- H. LEE, S. Oh, H. Chae, **S. Park**, J. Jekal, S. Cho, I. Moon, T. Lee, Y. Lee, J. Lim, Y. Park, and K. Jang, "Transcutaneous laryngeal disorder diagnosing wearables and clinical validation", *Science Translational Medicine*, under review (2026) [IF: 15.6] (Q1, top 2% in Multiple)

Peer-Reviewed Journal Articles

- S. Park**, J. Lee, J. Park, and I. Moon, "AI-driven dual-mode phase and label-free fluorescence imaging platform using a single-shot Gabor hologram", *Microsystems & Nanoengineering*, 12(1), 311 (2026) [IF: 11.1] (Q1, ranked 1st in Instruments & Instrumentation)
- S. Park**, O. Jeong, and I. Moon, "A photon-limited federated learning framework for privacy-aware image classification", *Information Sciences*, 123111 (2026) [IF: 6.8] (Q1, top 7.9% in Computer Science, Information Systems)
- S. Park**, J. Park, Y. Kim, I. Moon, and B. Javidi, "Simple and practical single-shot digital holography based on an unsupervised diffusion model", *Engineering Applications of Artificial Intelligence*, 163, 112970 (2026) [IF: 8.0] (Q1, top 2.5% in Engineering, Multidisciplinary)
- O. Jeong, **S. Park**, and I. Moon, "Scalable neural cryptanalysis of block ciphers in federated attack environments", *Mathematics*, 14(2), 373 (2026) [IF: 2.2] (Q1, top 6% in Mathematics)
- S. Park**, J. Lee, J. Park, and I. Moon, "HoloFluoNet: Live cell imaging intelligence based on fused holography and fluorescence for virtual staining, cell segmentation, classification, and viability analysis", *Engineering Applications of Artificial Intelligence*, 161, 112323 (2025) [IF: 8.0] (Q1, top 2.5% in Engineering, Multidisciplinary)
- S. Park**, H. An, A. Rehman, and I. Moon, "A holographic sensor-integrated deep learning framework for noninvasive assessment of stored red blood cell quality", *Advanced Sensor Research*, 4(12), e00073 (2025) [IF: 5.0]
- E. Ahmadzadeh, **S. Park**, Y. Kim, and I. Moon, "Automated phenotypic analysis and classification of drug-treated cardiomyocytes via synergized time-lapse holographic imaging and deep learning", *Computer Methods and Programs in Biomedicine*, 269, 108890 (2025) [IF: 4.9] (Q1, top 13.5% in Computer Science, Theory & Methods)
- S. Park**, M. Y. Kim, J. Jeong, S. Yang, M. S. Kim, and I. Moon, "Quantitative analysis of the dexamethasone side effect on human-derived young and aged skeletal muscle by myotube and nuclei segmentation using deep learning", *Bioinformatics*, 41(1), btac658 (2025) [IF: 5.4] (Q1, top 9.8% in Mathematical & Computational Biology)
- S. Park**, Y. Kim, and I. Moon, "Automated phase reconstruction and super-resolution with deep learning in digital holography", *Optics & Laser Technology*, 176, 111030 (2024) [IF: 5.0] (Q1, top 17.5% in Optics)
- A. Rehman, H. An, **S. Park**, and I. Moon, "Automated classification of elliptical cancer cells with stain-free holographic imaging and self-supervised learning", *Optics & Laser Technology*, 174, 110646 (2024) [IF: 5.0] (Q1, top 17.5% in Optics)
- S. Park**, Y. Kim, and I. Moon, "Fast automated quantitative phase reconstruction in digital holography with unsupervised deep learning", *Optics and Lasers in Engineering*, 167, 107624 (2023) [IF: 5.6] (Q1, top 17.3% in Optics)
- S. Park**, H. Kim, and I. Moon, "Automated classical cipher emulation attacks via unified unsupervised generative adversarial networks", *Cryptography*, 7(3), 35 (2023) [IF: 1.6] (Q2 in Computer Science, Theory & Methods)
- K. Jaferzadeh, S. Son, A. Rehman, **S. Park**, and I. Moon, "Automated stain-free holographic image-based phenotypic classification of elliptical cancer cells", *Advanced Photonics Research*, 4(1), 2200043 (2023) [IF: 3.7] (Q2 in Optics)
- E. Ahmadzadeh, K. Jaferzadeh, **S. Park**, S. Son, and I. Moon, "Automated analysis of human cardiomyocyte dynamics with holographic image-

based tracking for cardiotoxicity screening", *Biosensors and Bioelectronics*, 195, 113570 (2022) [IF: 12.6] (Q1, top 2.8% in Chemistry, Analytical)

3. E. Kim, **S. Park**, S. Hwang, I. Moon, and B. Javidi, "Deep learning-based phenotypic assessment of red cell storage lesions for safe transfusions", *IEEE Journal of Biomedical and Health Informatics*, 26(3), 1318–1328 (2022) [IF: 7.0] (Q1, top 6.2% in Mathematical & Computational Biology)
2. **S. Park**, Y. Kim, and I. Moon, "Automated phase unwrapping in digital holography with deep learning", *Biomedical Optics Express*, 12(11), 7064–7081 (2021) [IF: 3.9] (Q1, top 17% in Optics)
1. F. Yi, **S. Park**, and I. Moon, "High-throughput label-free cell detection and counting from diffraction patterns with deep fully convolutional neural networks", *Journal of Biomedical Optics*, 26(3), 036001 (2021) [IF: 2.7] (Q2 in Optics)

International Conference Papers and Abstracts

8. T. Bui, **S. Park**, L. El Zahar, and I. Moon, "Cycle-consistent diffusion modeling for bidirectional cryptanalysis of double random phase encoding in federated settings", *Proc. SPIE 14033, Three-Dimensional Imaging, Visualization, and Display 2026*, 81 (2026)
7. H. Woo, **S. Park**, J. Lee, I. Moon, and M. S. Kim, "A lightweight self-supervised deep learning framework for automated detection of circulating tumor cells and cancer-associated fibroblasts", *Cancer Research*, 86(7_Supplement), 119 (2026) [AACR Annual Meeting abstract]
6. **S. Park** and I. Moon, "Automated photon-counted image reconstruction using GANs", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025) — **Best Conference Paper Award**
5. A. D. Martin, O. Jeong, **S. Park**, and I. Moon, "Deep learning-based predictive models over pseudo random number generators", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025)
4. L. El Zahar, **S. Park**, and I. Moon, "Generative adversarial network-based cryptanalysis of double random phase encoding schemes", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025)
3. **S. Park** and I. Moon, "Photon-counting imaging with denoising diffusion models", *Proc. SPIE, Optics + Photonics*, San Diego, CA, USA (2025)
2. L. El Zahar, **S. Park**, and I. Moon, "Double random phase-encoded image reconstruction based on denoising diffusion models", *Proc. SPIE, Optics + Photonics*, San Diego, CA, USA (2025)
1. **S. Park**, J. Jeong, and I. Moon, "Automatic classification of cancer cells using digital holography", *Optica Advanced Photonics Congress — Novel Optical Materials and Applications*, NoTu2C.7 (2023)

Domestic Conference Papers

1. **S. Park** and I. Moon, "Unsupervised GAN-based substitution cipher attack model using imbalanced datasets", *Proceedings of the Korea Computer Congress (KIIE)*, 882–884 (2022) [in Korean]

PATENTS AND SOFTWARE REGISTRATIONS

Registered Patents — Republic of Korea

3. I. Moon, **S. Park**, "Deep learning-based apparatus and method for evaluating red blood cells". Reg. No. 10-2761648, registered 23 Jan. 2025 (filed 10-2022-0101534, 12 Aug. 2022)
2. I. Moon, H. Kim, **S. Park**, O. Jeong, "Apparatus and method for cryptanalysis using generative adversarial networks". Reg. No. 10-2522109, registered 11 Apr. 2023 (filed 10-2020-0144471, 2 Nov. 2020)
1. I. Moon, **S. Park**, Y. Kim, "Deep learning-based apparatus and method for phase unwrapping of digital holographic phase images". Reg. No. 10-2512155, registered 16 Mar. 2023 (filed 10-2020-0085656, 10 Jul. 2020)

Pending Patent Applications — PCT (International)

3. I. Moon, **S. Park**, Y. Kim, "Method and system for automated cryptanalysis of block and public-key ciphers based on distributed deep learning". PCT/KR2026/008776, filed 8 Jul. 2026
2. I. Moon, **S. Park**, "Cell analysis system and method based on virtual fluorescence generation". PCT/KR2025/004767, filed 8 Apr. 2025
1. I. Moon, **S. Park**, "Unsupervised learning-based method and system for phase image reconstruction using Gabor holograms". PCT/KR2025/004263, filed 1 Apr. 2025

Pending Patent Applications — Republic of Korea

6. I. Moon, **S. Park**, "Apparatus and method for dual-mode image generation and analysis based on a single-shot Gabor hologram". App. No. 10-2026-0127425, filed 10 Jul. 2026
5. I. Moon, J. Park, **S. Park**, Y. Kim, "Apparatus and method for privacy-preserving federated learning using photon-limited images". App. No. 10-2026-0127418, filed 10 Jul. 2026
4. I. Moon, **S. Park**, Y. Kim, "Method and system for automated cryptanalysis of block and public-key ciphers based on distributed deep learning". App. No. 10-2025-0139155, filed 25 Sep. 2025
3. I. Moon, **S. Park**, "Cell analysis system and method based on virtual fluorescence generation". App. No. 10-2024-0157925, filed 8 Nov. 2024
2. I. Moon, **S. Park**, "Unsupervised learning-based system and method for phase image reconstruction using Gabor holograms". App. No. 10-2024-0153232, filed 1 Nov. 2024
1. I. Moon, **S. Park**, "Method and system for drug screening of cardiomyocytes using holographic images". App. No. 10-2023-0140106, filed 19 Oct. 2023

Registered Software Copyrights — Republic of Korea

5. I. Moon, **S. Park**, "Target rare cell identification program based on bright-field images". Reg. No. C-2026-037293, 30 Jul. 2026

4. I. Moon, S. Park, "Optical cryptosystem security analysis program based on adversarial generative models". Reg. No. C-2025-043253, 21 Oct. 2025

3. I. Moon, S. Park, "Photon-counting image reconstruction program based on adversarial generative models". Reg. No. C-2025-043254, 21 Oct. 2025

2. I. Moon, S. Park, "Live cell analysis program based on virtual fluorescence generation". Reg. No. C-2024-037615, 17 Oct. 2024

1. I. Moon, H. Kim, S. Park, O. Jeong, "Automated classical cipher analysis program based on generative adversarial networks". Reg. No. C-2020-035557, 15 Oct. 2020

RESEARCH PROJECTS

AI-Driven Multi-parametric Liquid Biopsy Platform for Precision Tumor and Tumor Microenvironment Diagnostics

– Nov. 2025 – Present · Participating researcher — label-free detection and classification of circulating tumor cells and cancer-associated fibroblasts.

Development of Platforms for Analyzing Cellular Phenotypes based on AI and Multi-mode Bio-imaging

– Apr. 2023 – Present · Participating researcher — dual-mode holography/fluorescence imaging and virtual staining for live-cell phenotyping.

Research on AI-based Cryptanalysis and Security Evaluation

– Apr. 2020 – Dec. 2027 · Participating researcher — generative and federated deep learning for classical, block, and optical cryptosystem analysis.

HONORS AND AWARDS

Best Student Award, DGIST, Daegu, South Korea

– Awarded by the President of DGIST in recognition of outstanding research performance.

Best Conference Paper Award, ICEIB 2025, Taiwan

– 5th International Conference on Electronic Communications, Internet of Things and Big Data — for "Automated photon-counted image reconstruction using GANs".

TEACHING

Teaching Assistant, Optical Signal Processing, DGIST

Spring 2020

TECHNICAL SKILLS

Programming	Python, Bash
Deep learning	PyTorch; diffusion models, GANs, self-/unsupervised learning, federated learning, physics-informed networks, segmentation and classification
Optical imaging	Digital holography (Gabor, off-axis), quantitative phase imaging, fluorescence and confocal microscopy
Experimental	Mammalian cell culture, live-cell time-lapse imaging, fluorescence staining, sample preparation and optical bench alignment for holographic imaging
Systems	Linux server administration, multi-GPU training, Git
Languages	Korean (Native), English (Academic Proficiency)